Supplement to: Layouts Made in Diagram Editors Are Held by Alignment and Flow, Not by Edge Length or Stress

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The sensitivities, controls and secondary estimands of the paper, one section each (S1 to S8), with the script that produced every table named beside it. Section numbers cited as "Section n" are the paper's. The corpora, the terms and the test are as the paper defines them. Every BPMN number is on the seeded random 300 except where a line says first 300: the border-to-border run of S3, the pitch and specification-grid lines of S2, the pitch-5 line and the constrained-model probe of the paper's Sections 5 and 4, and the 41 to 100 band, which predate the sample and were not rerun.

S1 The summed energies, `prism` and the smooth kernel

$q_{0.02}$ under the summed energies at equal weights, with the `prism` column and the A3 row the main table omits. `prism` differs from `neato` under overlap alone, which it removes (1.00 at value 0.000 in every corpus), and under stress, where the removal costs it 0.06 to 0.11 of the hold.

corpus	energy	hand	`neato`	`prism`	`dot`	ELK
WikiPathways	alignment A3 alone	0.20 [0.18, 0.23] (0.284)	0.03 (0.446)	0.03 (0.451)	0.45 (0.204)	0.35 (0.176)
	C+O	1.00 [1.00, 1.00]	0.60	1.00	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.17	0.19	0.00	0.00
	C+O+S	0.00 [0.00, 0.00]	0.60	0.94	0.00	0.00
	C+O+R	0.21 [0.19, 0.23]	0.00	0.00	0.19	0.22
	C+O+A1	0.51 [0.47, 0.53]	0.04	0.07	1.00	0.57
	C+O+A3	0.24 [0.22, 0.27]	0.03	0.03	0.50	0.41
	C+O+N	0.86 [0.83, 0.88]	0.55	1.00	0.78	0.74
	C+O+F	0.56 [0.53, 0.59]	0.30	0.50	1.00	1.00
Reactome	alignment A3 alone	0.10 [0.08, 0.12] (0.300)	0.03 (0.430)	0.03 (0.428)	0.39 (0.206)	0.31 (0.213)
	C+O	1.00 [1.00, 1.00]	0.76	1.00	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.33	0.27	0.00	0.00
	C+O+S	0.00 [0.00, 0.00]	0.76	0.89	0.00	0.00
	C+O+R	0.17 [0.16, 0.19]	0.00	0.00	0.14	0.13
	C+O+A1	0.22 [0.19, 0.25]	0.05	0.07	1.00	0.43
	C+O+A3	0.11 [0.09, 0.12]	0.03	0.03	0.42	0.33
	C+O+N	0.94 [0.91, 0.95]	0.71	1.00	0.80	0.75
	C+O+F	0.61 [0.58, 0.65]	0.33	0.44	1.00	1.00
BPMN	alignment A3 alone	0.33 [0.28, 0.35] (0.213)	0.00 (0.454)	0.00 (0.453)	0.21 (0.187)	0.22 (0.163)
	C+O	1.00 [1.00, 1.00]	0.41	1.00	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.21	0.28	0.00	0.00
	C+O+S	0.00 [0.00, 0.00]	0.44	0.93	0.00	0.00
	C+O+R	0.74 [0.71, 0.76]	0.00	0.00	0.29	0.39
	C+O+A1	0.85 [0.82, 0.89]	0.04	0.09	0.95	0.67
	C+O+A3	0.35 [0.31, 0.38]	0.03	0.03	0.33	0.41
	C+O+N	1.00 [1.00, 1.00]	0.40	1.00	0.86	0.79
	C+O+F	0.86 [0.82, 0.88]	0.25	0.61	1.00	1.00

*Independent researcher. Code, parsed corpora and every table's script: `https://github.com/Anode1/cjitter`, directory `example/diagrams`.

At the hand layout the three base terms share the energy as length 0.99, crossings 0.01, overlap 0.00, and $C{+}O{+}L$ inherits the length term's zero.

The four-term fit of Section 4 under the smooth kernel (`results/a3_fit.md`, `fit.py` over the A3 differences regenerated 5 September 2026), beside the corner kernel's:

corpus	kernel	orthogonality	alignment	node–edge	flow	residual	held out
WikiPathways	A1	0.031	0.950	0.002	0.017	0.0010	0.248 [0.236, 0.260]
	A3	0.315	0.082	0.028	0.575	0.0046	0.221 [0.210, 0.232]
Reactome	A1	0.017	0.923	0.036	0.024	0.0014	0.149 [0.142, 0.157]
	A3	0.000	0.000	1.000	0.000	0.0033	0.350 [0.097, 0.901]
BPMN	A1	0.066	0.881	0.008	0.045	0.0003	0.651 [0.617, 0.689]
	A3	0.250	0.043	0.142	0.565	0.0010	0.633 [0.530, 0.984]

S2 Radius and directions

q_d at $d \in \{0.005, 0.01, 0.02, 0.05\}$, hand layouts and `neato`'s (`profile.py -sweep`).

	WikiPathways				Reactome				BPMN			
energy	.005	.01	.02	.05	.005	.01	.02	.05	.005	.01	.02	.05
overlap, hand	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
length, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
stress, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
alignment A1, hand	0.52	0.54	0.52	0.44	0.12	0.17	0.21	0.19	0.88	0.88	0.85	0.81
gridiness, hand	0.94	0.91	0.87	0.75	0.82	0.75	0.65	0.57	1.00	1.00	0.95	0.87
flow, hand	0.59	0.59	0.59	0.59	0.62	0.62	0.62	0.63	0.85	0.85	0.86	0.86
length, `neato`	0.04	0.12	0.24	0.47	0.07	0.19	0.35	0.58	0.07	0.20	0.40	0.70
stress, `neato`	0.90	0.96	1.00	1.00	0.96	1.00	1.00	1.00	0.94	0.97	1.00	1.00
alignment A1, `neato`	0.04	0.05	0.06	0.06	0.03	0.05	0.06	0.07	0.03	0.04	0.06	0.06

The hand verdicts under overlap, length and stress do not move with the radius. `neato`'s under length do. A layout that is nearly a minimum is held at a large radius and free at a small one, and `neato`'s layouts, which minimise stress and not length uniformity, read as nearly uniform at $d = 0.05$ and not at $d = 0.005$. A hand layout at 0.00 across the range is far from the minimum at every scale the test resolves. At d equal to one grid pitch, 0.0008 to 0.0017 of the width, hand length and stress still hold 0.00 (`results/pitch.md`, first 300).

Each doubling of V contains the previous set, so q can only fall as directions are added and the sixteen-direction figures are upper bounds on the continuum value. At 8, 16, 32 and 64 directions hand length and stress read 0.00 and `neato`'s stress 1.00 in every corpus. The A1 level falls: 0.58, 0.52, 0.47, 0.39 on WikiPathways, 0.31, 0.21, 0.14, 0.07 on Reactome and 0.88, 0.85, 0.82, 0.82 on BPMN. `neato`'s 0.06 falls to 0.00, and the random layouts' 0.06 to 0.07 falls to 0.03, 0.03 and 0.00 (`results/null64.md`). This is the corner's release bound narrowing. The bound: a box offset by ϵ from its nearest agreement is released only by a move whose component toward the agreement, a, is less than 2ϵ, since the move lands at $|\epsilon - a|$. The smallest such component among sixteen directions is $\sin 22.5° = 0.38$ of d, so release begins at $\epsilon > 0.19\,d$, 3 to 5 corpus units at $d = 0.02$. Each added direction lowers the smallest component and with it the bound. The limit lies below the exact-tie fraction, not at it: A1 sums over nodes, so a move that lowers an untied neighbour's distance releases a tied box, and only a layout where every box is tied, `dot`'s, keeps 1.00 at every count. The gap to the control survives at 64; levels are stated at sixteen. The full grid of radius, direction count, reference-length rule and route treatment, 1,008 cells, flips no verdict (`results/specgrid.md`, first 300).

S3 Reference length, box extent and routes

Under the median edge length instead of the fitted L, `neato`'s stress layouts score 0.07, 0.08 and 0.12 under stress and the hand layouts 0.00; under length the hand layouts score 0.03, 0.04 and 0.05 and `neato`'s 0.26, 0.41 and 0.43. Under $L = 1/\sqrt{N}$ no box of any layout is held under either term.

With distances measured border to border, each box's offset to its border along the joining direction subtracted and L refitted on those distances, length and stress hold 0.00 of the hand layouts in every corpus, 0.86 to 0.92 of diagrams at exactly zero (`results/border.py`, first 300). `neato`, which minimised the centre form, is not held under the border form (0.00 to 0.09).

Routes (`results/routes.md`): an edge is the polyline the file stores, attachment points and interior waypoints, or the chord between the centres where the file stores nothing. When a node moves its attachment points move with it and the waypoints stay. A waypoint is stored on 0.30, 0.07 and 0.34 of the edges by corpus, and only the BPMN routes are orthogonal, 0.87 axis-aligned at every segment against 0.01 and 0.00 in the biological formats. On chords (`data/*_chords.txt`) no verdict moves (`results/chords.md`).

S4 Descent, chance and calibration

The secondary estimand: hill climbing from the hand layout, each proposal one box moved a random step, kept when the energy falls, every box confined to a circle of radius 0.02 about its start, 15 seeds, against uniform sampling in the same circles at the same budget. Per corpus and energy: the median fraction of the cap the climber uses and of the energy it removes, the same for the sampler, the share of diagrams where the climber ends lower on at least 12 of 15 seeds, and $q_{0.02}$ where the uncapped climber stops, the converged reference.

		cap used		energy removed			
corpus	energy	climb	random	climb	random	$\geq$12/15	q conv.
WikiPathways	C+O+L	0.99	0.76	0.28	0.16	0.88	1.00
	C+O+A1	0.39	0.74	0.71	0.17	0.75	1.00
	stress	1.00	0.77	0.12	0.05	1.00	1.00
Reactome	C+O+L	1.00	0.76	0.31	0.16	0.93	1.00
	C+O+A1	0.56	0.74	0.99	0.37	0.80	1.00
	stress	1.00	0.77	0.13	0.06	1.00	1.00
BPMN	C+O+L	1.00	0.77	0.28	0.14	0.99	1.00
	C+O+A1	0.12	0.74	0.73	−0.01	0.95	1.00
	stress	1.00	0.77	0.15	0.07	1.00	1.00

Under the distance energies the climber walks every box to the cap circle and is still removing energy when the budget ends; under C+O+A1 it moves a fraction of the cap, the misalignments being nearer than d. The fraction of diagrams whose converged reference stops below 0.9 is at most 0.04 for the energies of the table, and below the plan's 0.95 at most 0.07, with 0.05 to 0.13 below 0.9 under $C{+}O{+}F$, the flow term's plateaus (`results/desc.md`).

Uniform sampling at the same budget is the control the descents are read against, the matched-budget control of hyperparameter search [1] prescribed for genetic algorithms in 1995 [2].

Calibration (`results/calib.py`, `results/calib_values.py`) reports three quantities on `neato`'s converged layouts after every box is displaced by ϵ of the width in a seeded direction: the median fraction ρ of the stress energy the same capped climber removes, the median stress value, and $q_{0.02}$ under stress. The last says how far a converged layout can be displaced before the test stops holding it. ρ is not monotone in ϵ, the cap seeing a shrinking share of a growing energy beyond 0.05, so the hand layouts' ρ of 0.12 to 0.15 has two preimages and is not a distance. The

value is a distance, and by it the hand layouts sit beyond the 0.10 displacement and under half the random layouts.

		$\epsilon = 0$	0.005	0.01	0.02	0.05	0.1	hand	random
WikiPathways	ρ	0.001	0.007	0.024	0.087	0.245	0.213	0.116	
	stress	0.042	0.042	0.043	0.046	0.069	0.121	0.217	0.435
	$q_{0.02}$	1.00	0.94	0.36	0.06	0.00	0.00	0.00	0.00
Reactome	ρ	0.001	0.009	0.033	0.116	0.289	0.234	0.128	
	stress	0.034	0.034	0.035	0.039	0.065	0.118	0.212	0.469
	$q_{0.02}$	1.00	0.96	0.39	0.06	0.00	0.00	0.00	0.00
BPMN	ρ	0.001	0.025	0.089	0.276	0.436	0.263	0.151	
	stress	0.016	0.017	0.018	0.024	0.060	0.129	0.197	0.507
	$q_{0.02}$	1.00	0.94	0.35	0.05	0.00	0.00	0.00	0.00

S5 The paired tests

Per corpus, hand against `dot` on each criterion alone, the Hodges–Lehmann shift of $q_{0.02}$ with its 95% interval, zeros dropped, two-sided, Holm within the corpus (`results/tests/`). The plan had named `neato` as the paired control and a one-sided test. Against `neato` the shifts are −0.25, −0.37 and −0.39 under length, −0.98, −1.00 and −0.98 under stress and +0.46, +0.17 and +0.76 under alignment, every interval excluding zero, which the main table already shows. So the paired table is reported against `dot`, the control the hand layouts resemble. The shifts under crossings, overlap, length and stress are confined to the minority of diagrams where the pair differs at all. The shifts that matter are the profile's gaps: the hand layout sits below `dot` on alignment (−0.46 [−0.49, −0.43], −0.70 [−0.73, −0.67] and −0.05 [−0.10, −0.00]) and on flow (−0.40, −0.33 and −0.17), and above it on node–edge separation (+0.13, +0.18 and +0.25) and, in BPMN, on orthogonality (+0.41). The last two read `dot` on chords and the hand on its routes, so they compare the measurement as much as the tool.

S6 The BPMN samples

Of the archive's 29,810 files, 21,784 parse as models with ten or more nodes. Of those, 13,435 have every edge a BPMN flow or association and the rest are EPC, Petri net and UML models, and 6,723 of the 13,435 have a largest component of 15 to 40 nodes (`results/bpmn_selection.md`). The study was planned on the first 300 models by id. On all 6,723 models in the band the alignment hold is 0.87 (`data/bpmn_population_a1.csv`, regenerated from the archive by the commands in its header), against 0.91 on the first 300. Resampling 300 of the population 2,000 times puts 0.91 at the 99th percentile, outside the interval [0.83, 0.90] a random 300 gives. Every other term is where it was (`results/bpmn_population.md`). The seeded random 300 (`random.Random(1)`, `data/bpmnr.txt`) with all four tool controls regenerated is the corpus of every BPMN number in this paper. Its A1 hold is 0.85 and every other energy is within 0.017 of the first 300. The one paired verdict that changes is hand against `dot` on alignment: parity on the first 300 and below on the random 300, as in the other corpora (`results/bpmn_random300.md`).

S7 Corpus checks

The 41 to 100 band (108, 265 and 300 diagrams, the BPMN 300 being the first of the band by id; against `neato`; `results/band41.md`): length and stress hold 0.00 by hand in every corpus and `neato`'s stress 1.00; overlap 1.00; alignment 0.39, 0.16 and 0.74 against `neato`'s 0.07 to 0.08. Larger diagrams carry less alignment, and crossings stop holding every biological box (0.97 and 0.98).

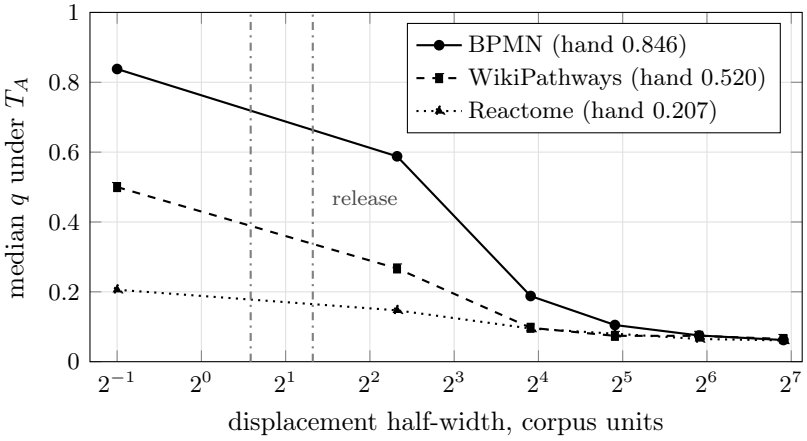


Figure 1: The displacement sweep, every box of every hand layout displaced by up to the half-width, the alignment hold read at $d = 0.02$ and sixteen directions. Two boxes displaced independently can be offset by twice the half-width, so the corner of T_A releases a box once the half-width passes about $0.1\,d$, 1.5 units in the biological corpora and 2.5 in BPMN (the two lines). The declared half-pitch test sits at the left edge, 0.5 units, where nothing has changed; past the lines the hold falls, BPMN 0.84 to 0.59 by a half-width of 5, and by 120 units every corpus is at the grid-snapped level of 0.06 to 0.09.

The grid controls (`results/alignment_controls.md`, `results/bpmnr_controls.md`). The snapped random layout holds 0.065, 0.075 and 0.070 at seed 1 and 0.06 to 0.09 over three seeds. The hand layouts displaced by up to half a pitch hold 0.500, 0.206 and 0.838 against 0.520, 0.207 and 0.846; displaced by up to 15 units, 0.097, 0.095 and 0.188; by up to 120, 0.065, 0.061 and 0.062. Under a jitter of 0.0025 of the extent, 2 to 3 units, on chords, the alignment hold keeps 0.44, 0.19 and 0.74 (`results/nulls.py`), while orthogonality falls from 0.16 to 0.06 in WikiPathways and from 0.43 to 0.19 in BPMN. Orthogonality, not alignment, is the tie artefact.

Eras (`results/era.md`): the GD Collection holds A1 at 0.81 in 1998 to 2005, 0.60 in 2006 to 2014 and 0.95 in 2015 to 2024; WikiPathways holds 0.47, 0.52, 0.53 and 0.53 across four generations of pathway numbers, so no gradient follows PathVisio's releases.

Figure 1 is the displacement sweep.

S8 The plan and its outcomes

The plan of 22 August 2026 (`PLAN-STATIONARITY.md`, signed that day) fixed its hypotheses after a 60-diagram pilot the same morning, so they carried no risk once the pilot had run, and the paper does not claim a registration. Its hypotheses and what the run found:

	hypothesis	outcome
H1	under $C{+}O$ the hand layout is held, median q above 0.9 in every corpus	held: 1.00
H2	under T_L alone it is not, median below 0.2	held: 0.00; also true of every non-optimised layout
H3	no positive weight on T_L repairs $C{+}O{+}L$	held: the step of Section 4
H4	alignment holds more hand-placed boxes than orthogonality or node–edge separation, and more than the tool control	refuted on the held fraction: node–edge holds 0.88 to 1.00 against alignment's 0.21 to 0.85; held on the value against `neato` and on the hold against `neato`; below `dot`
H5	under stress alone the hand layout is held less than `neato`'s	held: 0.00 against 1.00

Departures from the plan, all recorded after the analyses they concern: the plan's primary alignment definition is gridiness, and the paper reports the gridiness value with the A1 hold beside it; the flow term, declared as an alternative, is measured as a criterion; the reading direction is fitted like L; ELK, logged as unavailable, was obtained and run. The paired control is `dot` rather than `neato` and the test two-sided rather than one-sided (S5); the multiplicity family is the terms within one corpus against one control; the BPMN sample is the random 300 of Section S6. The radius in box widths, the A2 alignment definition and containment as a hard constraint, all declared, were not run.

References

[1] J. Bergstra, Y. Bengio. Random Search for Hyper-Parameter Optimization. *Journal of Machine Learning Research* 13:281–305, 2012.

[2] A. Juels, M. Wattenberg. Stochastic Hillclimbing as a Baseline Method for Evaluating Genetic Algorithms. *Advances in Neural Information Processing Systems 8* (NIPS 1995), MIT Press, 430–436, 1996.